Vector Database & RAG

Week 1: Modern AI

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Session Topics

- ✔ What is a Vector Database?
- ✔ How Does a Vector Database Work?
- ✔ What is RAG? (Retrieval-Augmented Generation)
- ✔ How Does RAG Work?
- ✔ Applications of RAG
- ✔ Benefits & Limitations of RAG

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What is a Vector Database?

- A Vector Database is a database designed to store, manage, and search vector embeddings efficiently.
- Embeddings represent the semantic meaning of text, images, documents, or other data as numerical vectors.
- Instead of searching only for exact keywords, vector databases can find information based on similarity and meaning.
- They are commonly used in modern AI applications that need to retrieve relevant information quickly.
- Vector databases are an important component of Retrieval-Augmented Generation (RAG) systems.

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How Does a Vector Database Work?

A vector database typically follows these steps:

Step 1 — Prepare Data

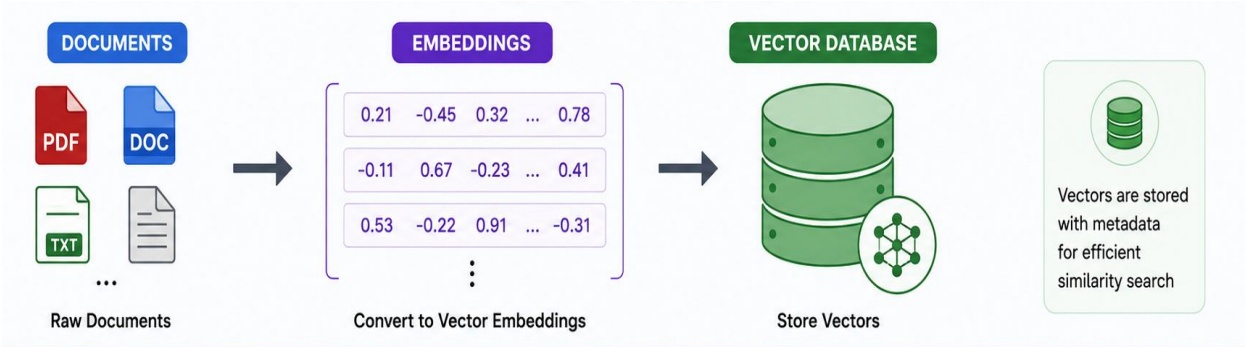
- Documents, text, images, or other information are collected and prepared.

Step 2 — Create Embeddings

- An embedding model converts the information into numerical vectors.

Step 3 — Store Vectors

- The vectors and relevant metadata are stored in the vector database.



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How Does a Vector Database Work?

Step 4 — Convert the Query

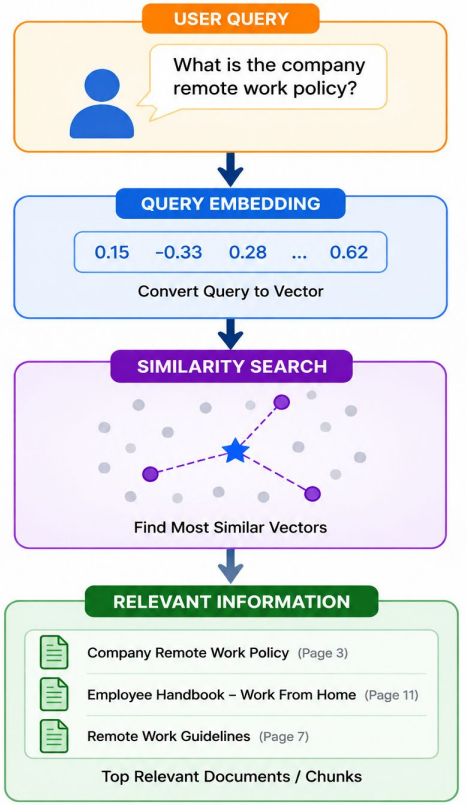
- When a user asks a question, the query is converted into an embedding using the same or compatible embedding model.

Step 5 — Similarity Search

- The database compares the query vector with stored vectors.
- It retrieves the vectors that are most similar to the query.

Step 6 — Return Relevant Information

- The retrieved documents or content are returned to the application for further processing.



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What is RAG(Retrieval-Augmented Generation)?

- RAG is an AI approach that combines information retrieval with Generative AI.
- Instead of relying only on the information learned during model training, the system retrieves relevant information from an external knowledge source.
- The retrieved information is provided to the LLM as additional context.
- The LLM then uses this context to generate a more relevant response.
- RAG is particularly useful when the required information is private, specialized, frequently changing, or not included in the model's training data.
- RAG allows an LLM to use external information when generating its response.

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How Does RAG Work?

A typical RAG system works in two main phases: Knowledge Preparation and Question Answering.

Phase 1 — Prepare the Knowledge Base

- Documents are divided into smaller chunks.
- Each chunk is converted into an embedding.
- The embeddings and associated text are stored in a vector database.

Phase 2 — Answer the User's Question

- The user's question is converted into an embedding.
- The vector database searches for similar content.
- The most relevant chunks are retrieved.

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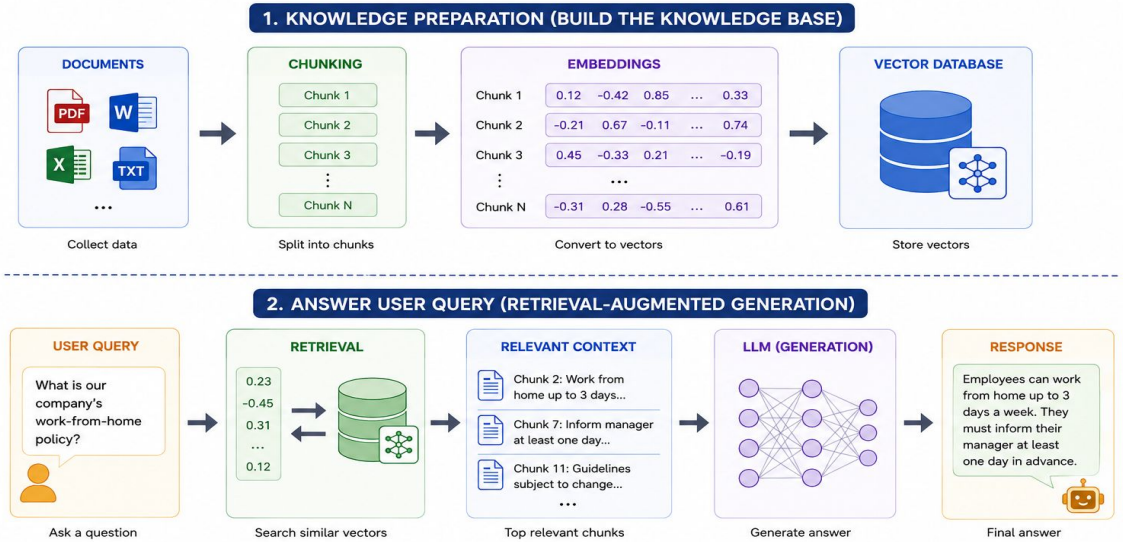
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How Does RAG Work?

Phase 3 — Generate the Response

- The retrieved information is provided to the LLM.
- The LLM uses the context to generate the response.



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Applications of RAG

- **Enterprise Knowledge Assistants** → Answer questions using internal company documents.
- **Customer Support** → Retrieve relevant product or support information.
- **Document Analysis** → Search and summarize large collections of documents.
- **Healthcare** → Retrieve relevant medical or organizational information.
- **Legal** → Search and retrieve relevant policies, contracts, or legal documents.
- **Education** → Build question-answering systems using course materials.

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Benefits & Limitations of RAG

Benefits

- **Uses Current Information** → Can retrieve information that changes over time.
- **Uses Private Knowledge** → Can work with organization-specific information.
- **Improves Relevance** → Provides the LLM with context related to the user's question.
- **Reduces Dependence on Model Memory** → Information can be retrieved from an external knowledge source.

Limitations

- **Retrieval Quality** → Poor retrieval can lead to poor answers.
- **Data Quality** → Outdated or incorrect documents can affect responses.
- **System Complexity** → RAG requires multiple components to work together.
- **LLM Errors** → The model may still generate incorrect information even when context is provided.

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